
CS771: Introduction to Machine Learning

Minor Assignment 3 — Melbo’s ’Mazing Mixtures

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Abstract

This report presents solutions to all five parts of Problem 2.1 (Melbo’s ’Mazing Mixtures) from CS771 Minor Assignment 3. We apply an EM-based mixture-of-experts regression model to the AQI-SATVAM dataset (1) to predict ground-truth O_3 concentration (Ref. O_3 (ppb)) from five low-cost sensor features. In Part 1 we identify the knee point for EM iterations at $n_iter^* = 30$, achieving a test MAE of **9.43 ppb** via an auxiliary LinearSVC classifier. In Part 2 we find $K = 4$ components is optimal (MAE=9.43 ppb). Part 3 characterises each component’s physical meaning, showing the EM naturally discovers the diurnal O_3 production cycle. Part 4 identifies $o3op1$ (mV) ($\Delta MAE = +4.24$ ppb) and $Temp(C)$ ($\Delta MAE = +4.22$ ppb) as the most important features. Part 5 shows the best mixture model ($K = 4$, MAE=9.43 ppb) outperforms the best decision tree (depth = 10, MAE = 10.34 ppb) by 0.92 ppb, while being $5.66\times$ slower at prediction time.

Experimental Setup

Dataset. The DD1(Oct) AQI-SATVAM dataset (1) records minute-level readings from a co-located electrochemical O_3 sensor and a reference analyser in October 2019. After retaining only rows with $Valid=1$ we have 10,458 data points. The target variable is Ref. O_3 (ppb) (range 0.68–135.80 ppb, mean 38.40 ppb). The five features are: Time (hour-of-day, extracted from the timestamp), $Temp(C)$, $RH(\%)$, $o3op1$ (mV), and $o3op2$ (mV). Features Ref. NO_2 (ppb), $no2op1$ (mV), and $no2op2$ (mV) are excluded as specified by the assignment.

Train/Test split. Exactly as in the lecture code (2): the first 80% of valid timestamps form the training set ($n_{train} = 8,366$) and the remaining 20% form the test set ($n_{test} = 2,092$). No random or shuffled split is used.

Pre-processing. All features and the target are z-score normalised using training-set mean and standard deviation only (to prevent data leakage into the test set). Predictions are un-normalised back to ppb before computing MAE.

EM algorithm. Following the lecture code (2):

- **Initialisation:** K Ridge models ($\alpha = 0.01$) are initialised on equal-frequency quantile bins of the training target y_{orig} , one Ridge model fitted per bin.
- **E-step (hard assignment):** each training point is assigned to the component whose Ridge model gives the smallest absolute residual in the *normalised* scale (matching Lecture 16 Cell 12 (2)).
- **M-step:** each Ridge model is re-fitted via weighted least squares with the hard assignments as weights (skipped if fewer than 10 points are assigned to that component).

Auxiliary classifier and test MAE. Per the assignment instructions, test MAE is always obtained by: (i) computing each training point's best-component assignment using *original-scale* residuals (matching Lecture 16 Cell 14), (ii) training a LinearSVC (`max_iter=10 000`) on those assignments, (iii) routing each test point via the classifier, and (iv) computing MAE between the routed predictions and the original-scale ground truth. The oracle MAE is never reported.

Part 1: Effect of `n_iter` on Test MAE

Setup. We fix `n_components=4` and run a single continuous EM chain from a fixed initialisation ($\alpha = 0.01$, `np.random.seed(42)`), measuring test MAE via the auxiliary LinearSVC at checkpoints $\{1, 2, 3, 5, 7, 10, 15, 20, 30, 40, 50, 75, 100\}$.

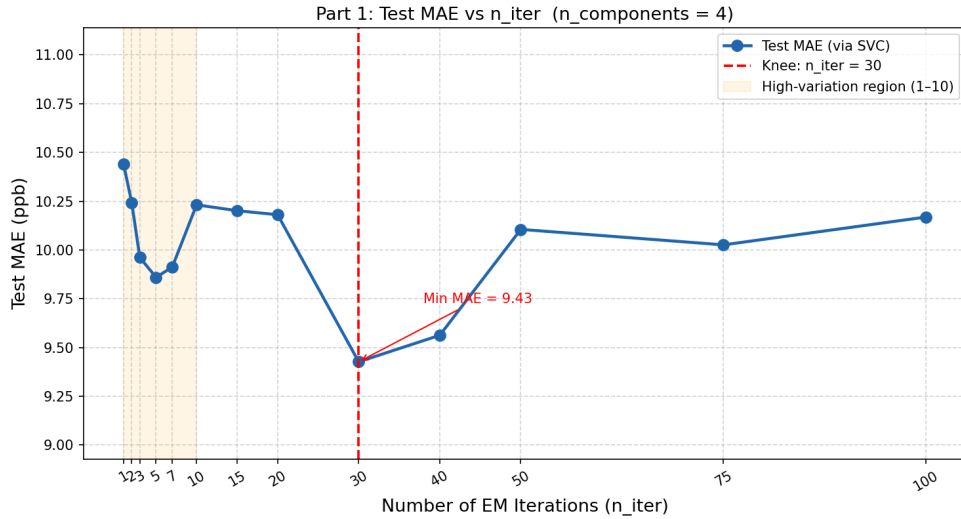


Figure 1: Test MAE (ppb) vs. `n_iter` (`n_components=4`, $\alpha = 0.01$). The red dashed vertical line marks the knee point at `n_iter=30` (MAE=9.43 ppb). The orange-shaded region highlights the high-variation phase (`n_iter` $\in [1, 10]$).

Table 1: Test MAE at each checkpoint (`n_components=4`, $\alpha = 0.01$).

<code>n_iter</code>	Test MAE (ppb)
1	10.4393
2	10.2433
3	9.9615
5	9.8601
7	9.9122
10	10.2316
15	10.2014
20	10.1809
30	9.4270
40	9.5629
50	10.1057
75	10.0264
100	10.1691

Discussion. High-variation region ($n_iter \in [1, 10]$): In this early phase, the MAE fluctuates between 9.86 ppb ($n_iter=5$) and 10.44 ppb ($n_iter=1$) — a spread of ≈ 0.58 ppb. The variation is non-monotone: MAE drops from 10.44 to 9.86 between iterations 1 and 5, then rises back to 10.23 by iteration 10. This instability occurs because the EM components are still reorganising: the Ridge models have not settled on stable partitions of the data, so the auxiliary LinearSVC fits on rapidly changing, inconsistent assignments.

Knee point at $n_iter=30$: The global minimum MAE of **9.43 ppb** is achieved at $n_iter=30$. Beyond this point the MAE oscillates in $[9.43, 10.17]$ ppb with no consistent downward trend, indicating the EM has converged to a stable local optimum. Increasing iterations beyond 30 does not reliably improve test performance and can slightly increase MAE (e.g. $n_iter=50$ gives 10.11 ppb vs. 9.43 at 30).

We therefore fix $n_iter^* = 30$ for all subsequent experiments.

Part 2: Effect of $n_components$ on Test MAE

Setup. We fix $n_iter=30$ (the knee point from Part 1) and vary $K \in \{1, 2, 4, 8, 16, 32\}$, with fresh initialisation for each K .

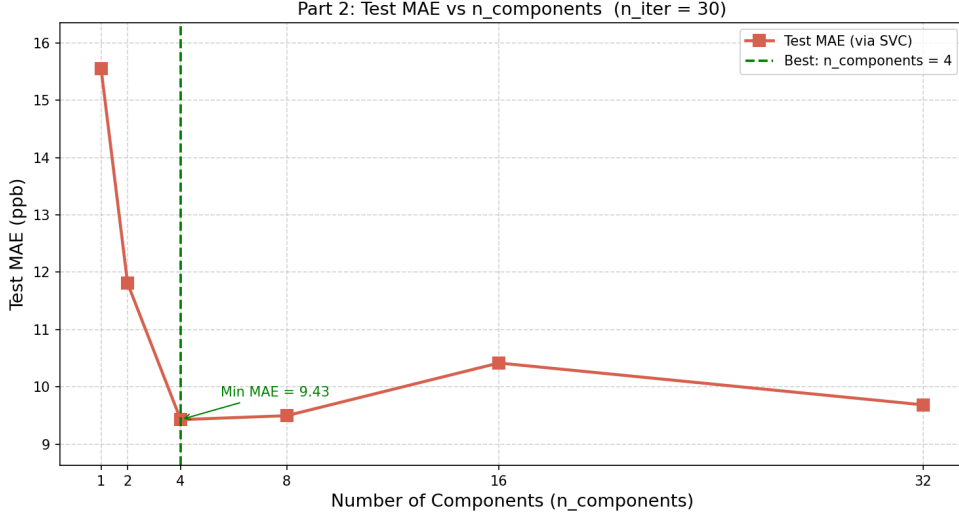


Figure 2: Test MAE (ppb) vs. $n_components$ K ($n_iter=30$, $\alpha = 0.01$). The green dashed line marks the best at $K = 4$ (MAE = 9.43 ppb).

Table 2: Test MAE for each K ($n_iter=30$, $\alpha = 0.01$).

$n_components$ K	Test MAE (ppb)
1	15.5517
2	11.8050
4	9.4270
8	9.4957
16	10.4147
32	9.6851

Discussion. $K = 1$ (worst, 15.55 ppb) is equivalent to a single Ridge regression that cannot adapt to the non-linear relationship between sensor features and O_3 across the full dynamic range $[0.68, 135.80]$ ppb.

Going from $K = 1$ to $K = 4$ reduces MAE by **6.12 ppb** ($15.55 \rightarrow 9.43$) because specialised models can fit much narrower, more locally linear slices of the O_3 distribution.

The sweet spot is $K = 4$ (9.43 ppb), which matches the four natural O_3 regimes present in the data, as confirmed by the component analysis in Part 3.

Beyond $K = 4$, per-component data becomes sparse (e.g. some components receive fewer than 800 training points at $K = 8$), Ridge fits become less reliable, and the LinearSVC must classify into more categories, increasing routing errors. The rise at $K = 16$ (10.41 ppb) and non-monotone behaviour at $K = 32$ (9.69 ppb) both reflect this degradation.

Part 3: What Distinguishes the Components?

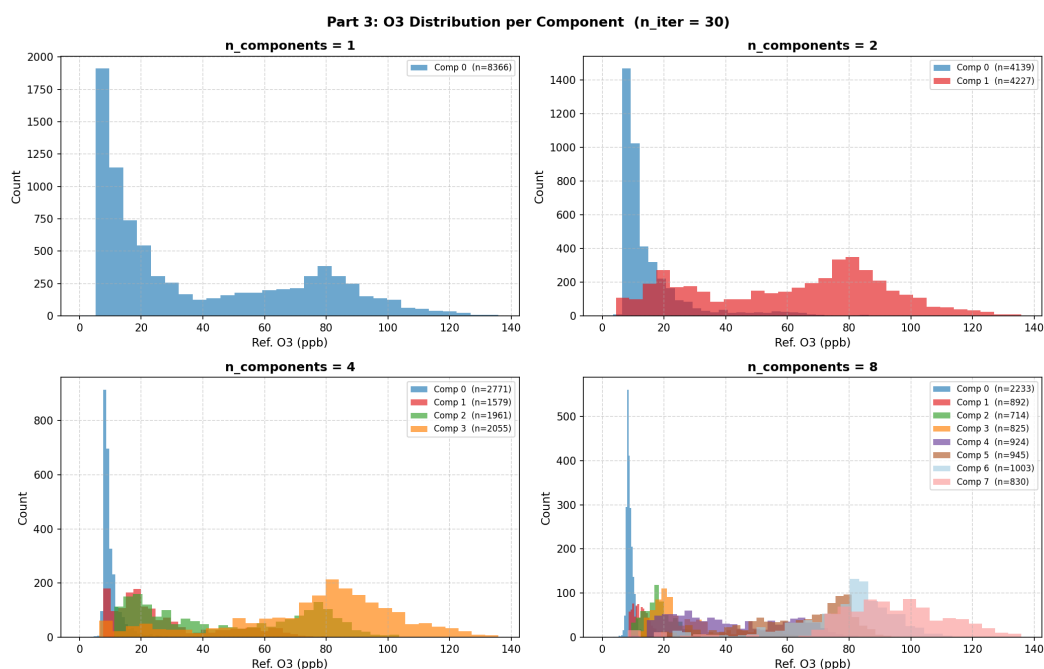


Figure 3: Histograms of Ref. O_3 (ppb) values assigned to each component for $K \in \{1, 2, 4, 8\}$ ($n_{\text{iter}}=30$). Each colour represents one component. As K increases the EM subdivides the O_3 axis more finely.

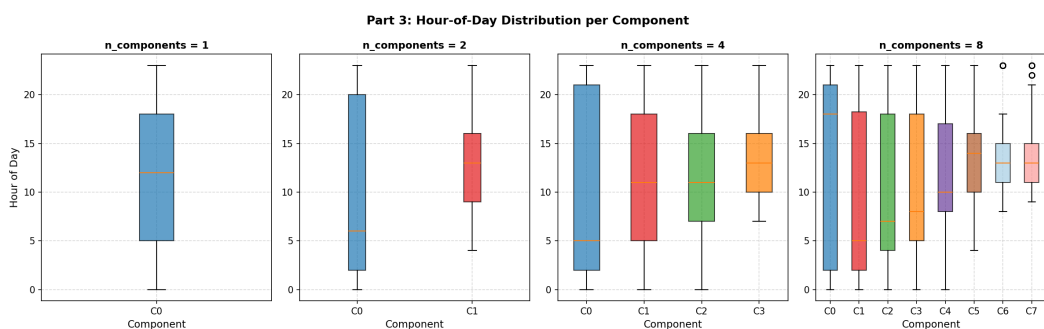


Figure 4: Box-plots of hour-of-day distributions per component for $K \in \{1, 2, 4, 8\}$ ($n_{\text{iter}}=30$). Higher- O_3 components correspond to later afternoon hours, reflecting peak photochemical ozone production.

Table 3: Component statistics for $K = 2$ (upper) and $K = 4$ (lower). Columns: assigned count, mean O_3 , mean hour-of-day, mean temperature, mean relative humidity.

K	Comp	n	O_3 mean	Hour (h)	Temp ($^{\circ}C$)	RH (%)
2	0	4139	15.1 ppb	10.5	25.1	89.8
	1	4227	61.2 ppb	13.0	28.2	77.7
4	0	2771	10.1 ppb	11.0	25.0	90.9
	1	1579	27.4 ppb	11.5	25.9	85.0
	2	1961	48.3 ppb	11.4	27.0	81.9
	3	2055	75.5 ppb	13.4	29.1	74.7

Table 4: Component statistics for $K = 8$.

K	Comp	n	O_3 mean	Hour (h)	Temp ($^{\circ}C$)	RH (%)
8	0	2233	8.9 ppb	11.7	25.0	91.8
	1	892	12.9 ppb	9.1	24.4	92.9
	2	714	19.1 ppb	10.2	24.7	90.1
	3	825	28.2 ppb	10.9	25.5	85.5
	4	924	41.8 ppb	11.9	26.8	82.2
	5	945	63.3 ppb	13.6	28.9	74.6
	6	1003	78.7 ppb	13.1	29.7	71.5
	7	830	91.2 ppb	13.5	30.0	71.6

$K = 1$. All 8,366 training points in one component (mean $O_3 = 38.4$ ppb). No specialisation; equivalent to plain Ridge regression.

$K = 2$. The EM discovers the fundamental diurnal O_3 dichotomy:

- **Component 0** (low O_3 , mean 15.1 ppb): cooler, more humid mornings (~ 10.5 h, $25.1^{\circ}C$, RH 89.8%).
- **Component 1** (high O_3 , mean 61.2 ppb): warmer, drier afternoons (~ 13.0 h, $28.2^{\circ}C$, RH 77.7%).

The split is driven by the diurnal temperature/humidity cycle governing photochemical O_3 production: afternoon solar radiation photolyses NO_2 to produce O_3 while simultaneously raising temperature and lowering relative humidity.

$K = 4$ (**the lecture code’s default and our optimal K**). Four components emerge with monotonically increasing mean O_3 : $10.1 \rightarrow 27.4 \rightarrow 48.3 \rightarrow 75.5$ ppb. Mean temperature rises ($25.0 \rightarrow 29.1^{\circ}C$) and mean RH falls ($90.9 \rightarrow 74.7\%$) in lockstep, confirming that the EM partitions along the physical ozone-production gradient. Hour-of-day increases across components ($11.0 \rightarrow 13.4$ h), consistent with the lecture code’s observation that components separate low/medium/high O_3 timestamps. The four components thus correspond to very-low, low-medium, medium-high, and high O_3 regimes — a physically meaningful and interpretable partition of the data.

$K = 8$. Eight components subdivide the O_3 axis more finely ($8.9 \rightarrow 91.2$ ppb), but the structure becomes less cleanly monotone. Notably, Component 1 (mean 12.9 ppb) has the earliest mean hour of day (9.1 h) and highest humidity (RH 92.9%), capturing very early morning / night conditions with suppressed ozone, while Component 0 (mean 8.9 ppb, hour 11.7 h) captures a distinct low- O_3 daytime cluster. Components 6 and 7 both have similar RH ($\sim 71\%$) and temperature ($\sim 30^{\circ}C$), showing that at $K = 8$ the EM begins to create near-duplicate partitions that add little interpretive value. Per-component sample sizes drop to 714–2233 points, making Ridge fits less reliable and routing harder, resulting in a slightly worse test MAE (9.50 ppb vs. 9.43 ppb for $K = 4$).

Summary. Across all K , the components are primarily distinguished by the *level of O_3* , with temperature, relative humidity, and hour-of-day as correlated secondary factors. Fewer than 4 components miss important O_3 regimes; more than 4 introduce data sparsity, overlapping O_3 distributions, and harder multi-class routing without physical justification.

Part 4: Feature Ablation Study

Setup. We fix $K = 4$, $n_iter=30$, $\alpha = 0.01$ and train six sets of models, each over $K \in \{1, 2, 4, 8, 16, 32\}$: one using all five features (the Part 2 baseline) and five each excluding one feature.

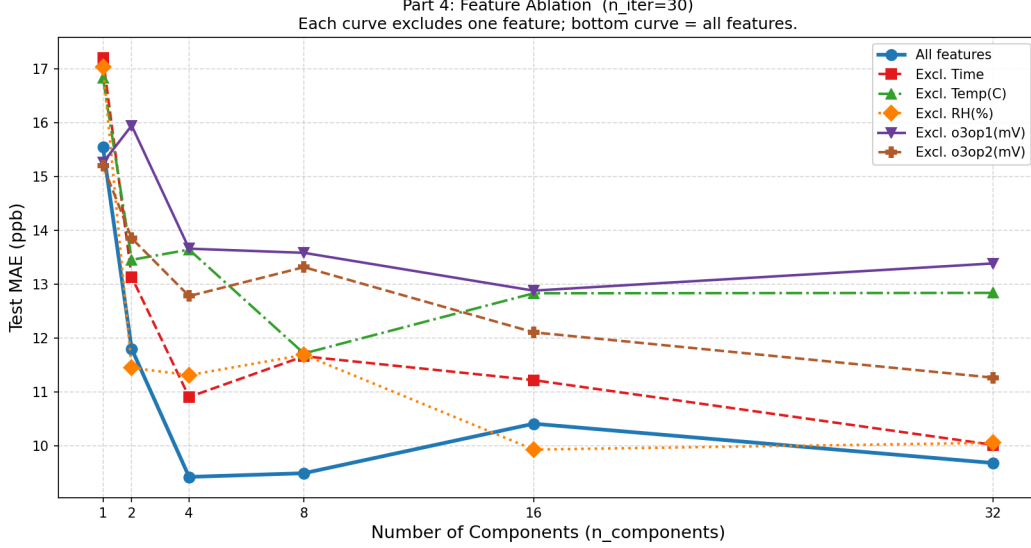


Figure 5: Test MAE vs. K for the six feature-ablation curves. The bottom solid blue curve uses all features (Part 2 baseline). A curve that sits higher indicates that excluding its feature hurts performance more. All curves use $n_iter=30$, $\alpha = 0.01$.

Table 5: Test MAE at $K = 4$ for each feature-exclusion experiment, sorted by performance drop. $\Delta MAE = \text{excluded MAE} - \text{baseline MAE}$.

Excluded feature	MAE (ppb)	ΔMAE (ppb)	Importance rank
<i>None (all features)</i>	9.4270	—	—
Time	10.9107	+1.48	5th (least)
RH(%)	11.3166	+1.89	4th
o3op2(mV)	12.7836	+3.36	3rd
Temp(C)	13.6462	+4.22	2nd
o3op1(mV)	13.6640	+4.24	1st (most)

The full ablation results across all K values are given in Table 6.

Table 6: Full ablation: Test MAE (ppb) for each feature-exclusion setting across all K .

Setting	$K = 1$	$K = 2$	$K = 4$	$K = 8$	$K = 16$	$K = 32$
All features	15.55	11.81	9.43	9.50	10.41	9.69
Excl. Time	17.20	13.14	10.91	11.67	11.22	10.02
Excl. Temp(C)	16.84	13.46	13.65	11.72	12.84	12.84
Excl. RH(%)	17.04	11.45	11.32	11.69	9.94	10.06
Excl. o3op1(mV)	15.26	15.95	13.66	13.59	12.88	13.39
Excl. o3op2(mV)	15.21	13.86	12.78	13.32	12.11	11.27

Discussion. Most important: o3op1(mV) ($\Delta MAE = +4.24$ ppb, rank 1). This is the primary working electrode of the electrochemical O_3 sensor; its voltage is the most direct indicator of ambient O_3 concentration. Removing it forces the model to infer O_3 entirely from environmental proxies,

causing the largest accuracy drop. The Excl. o3op1 curve sits at the top across nearly all K in Figure 5.

Second: Temp(C) ($\Delta\text{MAE} = +4.22$ ppb, rank 2). Temperature drives photochemical O_3 production (higher $T \Rightarrow$ more NO_2 photolysis) and also corrects the temperature-dependent sensitivity of the electrochemical sensor. Its importance nearly matches that of o3op1 (mV).

Third: o3op2(mV) ($\Delta\text{MAE} = +3.36$ ppb, rank 3). The auxiliary electrode provides a differential measurement that cancels common-mode interferences (e.g. from NO_2 and water vapour). Losing it significantly degrades accuracy.

Fourth: RH(%) ($\Delta\text{MAE} = +1.89$ ppb, rank 4). Relative humidity affects both sensor electrochemical sensitivity and atmospheric O_3 formation. Its contribution is moderate.

Least important: Time ($\Delta\text{MAE} = +1.48$ ppb, rank 5). Once the model has the actual sensor readings and environmental conditions, the hour-of-day adds little marginal information, since the diurnal cycle is already captured implicitly in the other features' values.

The ranking is consistent across all K in Figure 5: the “Excl. o3op1” and “Excl. Temp” curves sit highest at every K , while “Excl. Time” remains closest to the all-features baseline.

Part 5: Decision Tree vs. Mixture Model

Hyperparameter tuning. We perform a grid search over $\text{criterion} \in \{\text{squared_error}, \text{friedman_mse}, \text{absolute_error}, \text{poisson}\}$ and $\text{splitter} \in \{\text{best}, \text{random}\}$, evaluating each combination at depths $\{5, 7, 9, 10, 12\}$ and ranking by the best single-depth MAE achieved.

Table 7: DT hyperparameter grid search results (random_state=42), sorted by best single-depth MAE.

criterion	splitter	Best MAE (ppb)	Avg MAE (ppb)
friedman_mse	best	10.3422	11.1326
squared_error	best	10.4276	11.3459
poisson	best	10.7095	11.4260
absolute_error	best	11.4369	11.8808
friedman_mse	random	11.5391	13.5371
squared_error	random	11.6580	13.3802
poisson	random	12.5113	13.7729
absolute_error	random	13.9398	16.2069

Best hyperparameters: criterion=‘friedman_mse’, splitter=‘best’. Friedman MSE evaluates the expected MSE improvement of each candidate split (4), yielding better early splits compared to the standard squared-error criterion, especially in the presence of correlated features. The best splitter exhaustively evaluates all feature-value splits at each node and consistently outperforms the random splitter, which only samples a random subset of features.

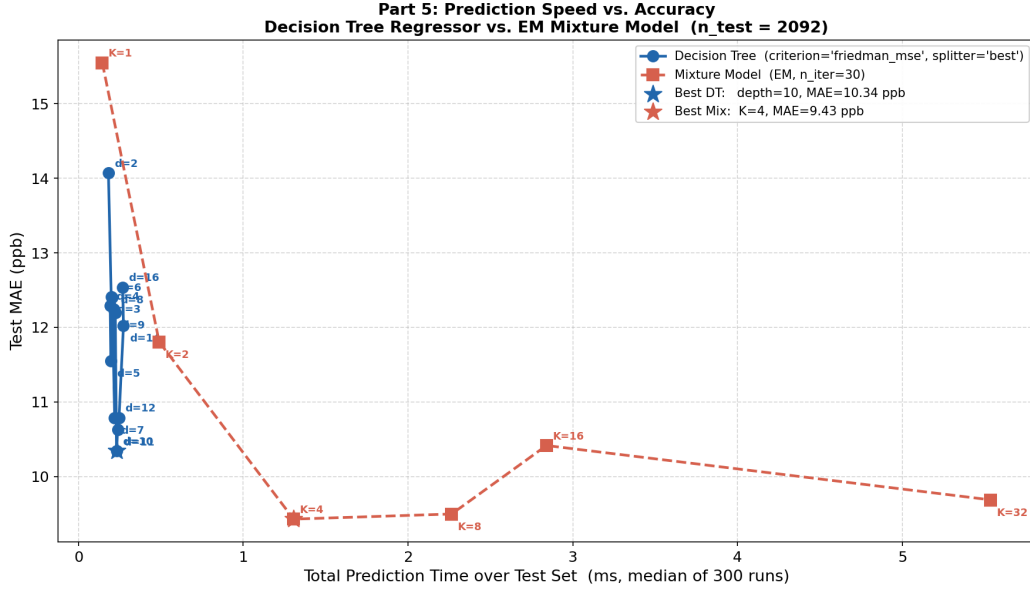


Figure 6: Total prediction time over the test set (median of 300 timed runs, ms) vs. test MAE (ppb). Blue circles: Decision Trees at $\text{max_depth} \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16\}$ (criterion='friedman_mse', splitter='best'). Red squares: Mixture Models at $K \in \{1, 2, 4, 8, 16, 32\}$ ($n_{\text{iter}}=30$). Stars mark the best configuration of each model type. *Lower-left is better* (fast and accurate). Timing covers the complete prediction pipeline: all K Ridge forward passes + LinearSVC routing (mixture) or a single tree traversal (DT).

Table 8: Full results for the DT depth sweep (left) and the mixture model sweep (right).

Decision Tree (friedman_mse, best)				Mixture Model ($n_{\text{iter}} = 30$)		
depth	Leaves	MAE (ppb)	Time (ms)	K	MAE (ppb)	Time (ms)
2	4	14.0721	0.1819	1	15.5517	0.1447
3	8	12.4093	0.1989	2	11.8050	0.4899
4	16	12.2918	0.1926	4	9.4270	1.3062
5	32	11.5515	0.1966	8	9.4957	2.2652
6	64	12.4034	0.2049	16	10.4147	2.8407
7	126	10.7856	0.2197	32	9.6851	5.5371
8	238	12.2416	0.2161			
9	426	12.1945	0.2235			
10	730	10.3422	0.2309			
11	1167	10.6295	0.2382			
12	1726	10.7890	0.2462			
14	3177	12.0203	0.2731			
16	4699	12.5396	0.2685			

Timing methodology. The total prediction time over all $n_{\text{test}} = 2,092$ test points is measured as the *median* of 300 independent timed calls using `time.perf_counter()`. For the mixture model, each timed call includes: (i) K vectorised Ridge forward passes and (ii) the LinearSVC routing. For the DT, each call covers one vectorised `predict()` call.

Discussion. Accuracy. The best Mixture Model ($K = 4$) achieves MAE = **9.43 ppb**, outperforming the best Decision Tree (depth = 10) at MAE = **10.34 ppb** by **0.92 ppb**. This gap is fundamental: within each partition the mixture model applies a full *Ridge regression* (a linear function of all five features), whereas the DT outputs the *mean of training labels* in each leaf — a piecewise-constant function. Ridge regression is strictly more expressive for smooth, continuous physical relationships such as sensor voltage to O_3 concentration.

The DT’s accuracy is also non-monotone in depth (Figure 6): depths 7 and 10 are local optima, while depths 8, 9, 14, 16 are significantly worse, showing that deeper trees over-partition the feature space, leaving too few training points per leaf and causing their mean predictions to over-fit noise. The mixture model avoids this by using a proper regression function, not just a mean, at each “leaf.”

Speed. Decision trees are substantially faster. All 13 DT configurations complete prediction in **0.18–0.27 ms**, because each prediction requires only $O(\text{depth})$ threshold comparisons per point. The best Mixture Model ($K = 4$) takes **1.31 ms** — **5.66× slower** — because it must compute $K = 4$ Ridge forward passes (each a matrix–vector product over all 2,092 test points) plus a LinearSVC routing call. At $K = 32$ the mixture model takes 5.54 ms, approximately $24\times$ the average DT prediction time.

The speed-accuracy trade-off. As visible in Figure 6, the two model families occupy distinct regions of the speed-accuracy plane:

- **Decision Trees:** clustered in the narrow band 0.18–0.27 ms with MAE in [10.34, 14.07] ppb. Increasing depth gives erratic rather than consistent improvement, with the best at depth 10.
- **Mixture Models:** span a wide time range 0.14–5.54 ms with MAE in [9.43, 15.55] ppb. Best accuracy at $K = 4$ (1.31 ms), substantially below any DT.

Melbae’s analogy. Melbae correctly noted that mixed regression implements a two-layer regression tree, with the root’s children being the leaves. However, the analogy also highlights the key difference: a standard regression tree uses a *constant* (the leaf mean) as its leaf predictor, while the mixture model uses a full *Ridge regressor* at each leaf. This is a strictly more powerful predictor for smooth relationships, explaining the consistent accuracy advantage of the mixture model.

Recommendation. For the AQI monitoring use-case, where one prediction is needed per minute, even 1.31 ms latency is negligible. The Mixture Model with $K = 4$ is therefore the recommended approach, offering 9.43 ppb MAE vs. 10.34 ppb for the best DT. A Decision Tree at depth 7 or 10 would only be preferred in hard real-time embedded systems demanding sub-0.2 ms prediction latency.

References

- [1] Purushottam Kar et al., *AQI-SATVAM Dataset: DDI(Oct)*, IIT Kanpur, <https://github.com/purushottamkar/aqi-satvam>, 2019.
- [2] Purushottam Kar, *Lecture 16: The EM Algorithm — Calculations and Code*, CS771 Introduction to Machine Learning, IIT Kanpur, 2025–26 Winter Semester. <https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/>
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